Prove closure rules:

if we have regular lang L, transform(L) is also regular.

For any regular language L, the language $L^R = \{w^R : w \in L\}$ is also regular.

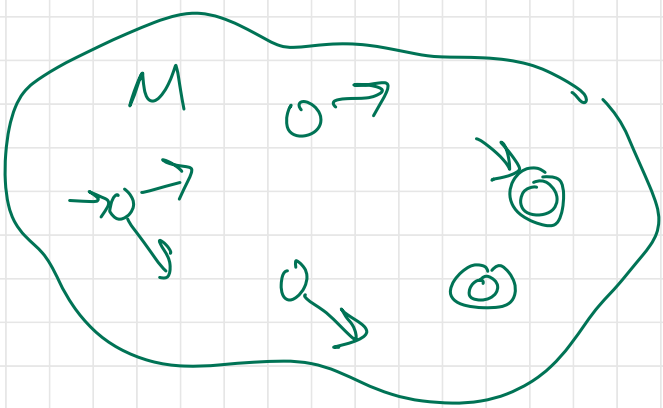
$L = 0^* 1^*$

$L^R = 1^* 0^*$

example

Proof

Let L be a regular language.
Let $M = (Q, s, A, \delta)$ be a DFA that accepts L.



how to transform M so that it accepts L^R?

reverse every edge ←

swap accepting, starting states

M^R:

$Q^R = Q$

$s^R = A$

$A^R = \{s\}$

$\delta^R(p,a) = \{q : p = \delta(q,a)\}$ ↙

$$q \xrightarrow{a} p \quad \Rightarrow \quad q \longleftarrow p$$

$\delta(q,a) = p$